

Bayesian Hypothesis test, Bayes factor and Information criteria

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5.3 Bayesian Information Criterion

- BIC can be viewed as a substitute to BF in model comparison

→ This is often used outside of a Bayesian framework b/c it doesn't require a prior distribution (so it really doesn't know anything to do w/ Bayes)

- BIC can compare more than two models

- BIC doesn't required priors

- Definition**

$$BIC = \underbrace{-2 \log L(\hat{\theta} | y)}_{\text{How well the model fits the observed data}} + \underbrace{p \log(n)}_{\text{penalty for size of model}}$$

– Good models have relatively small BIC values

- Relationship between BIC and BF: For large n

$$BF_{12} \approx e^{-\frac{1}{2}(BIC_{M_1} - BIC_{M_2})}$$

$$\begin{aligned} \rightarrow p(y|M) &= \int p(y|\theta, M) \pi(\theta|M) d\theta \quad \text{can be approximated by} \\ \rightarrow p(M|y) &= \frac{p(y|M)}{p(y)} = \frac{p(y|M)p(M)}{p(y)} \approx \frac{\exp\{-BIC_M/2\} p(M)}{p(y)} \quad \text{derivation} \\ \rightarrow \text{if } p(M_1) = p(M_2) &\Rightarrow BF_{12} = \frac{p(M_1|y)}{p(M_2|y)} \\ &\approx \frac{\exp\{-BIC_{M_1}/2\}}{\exp\{-BIC_{M_2}/2\}} \\ &= \exp\left\{-\frac{1}{2}(BIC_{M_1} - BIC_{M_2})\right\} \end{aligned}$$

5.4 Deviance and Deviance Information Criterion(DIC)

5.4.1 Deviance

→ Intuition: Deviance is kind of like a likelihood of the data, plus some constant. The constant is just the distribution of y , so we will focus only on the model part

- For a data distribution $y|\theta$:

$$\text{Deviance} = D(\theta) = -2 \log p(y|\theta) + C$$

→ This is just a fancy function of θ
→ we can find its expectation like below

- For example, $Y_1, Y_2, \dots, Y_n \sim$ independent $\text{Bin}(n_i, \theta_i)$, the deviance is

$$D(\theta) = -2 \log \prod_{i=1}^n \binom{n_i}{y_i} \theta_i^{y_i} (1 - \theta_i)^{n_i - y_i}$$

- Posterior mean of deviance**

$$\begin{aligned} E_{\text{post}}(D) &= E_{\theta|y}[D(\theta)] \\ &= -2E_{\theta|y}[\log p(y|\theta)] \end{aligned}$$

– $E(D)$ is invariant for parameterization

– Estimate of $E(D)$ is denoted as

$$\bar{D} = \frac{1}{S} \sum_{s=1}^S D(\theta^s) = -2 \frac{1}{S} \sum_{s=1}^S \log p(y|\theta^s),$$

where θ^s are from the posterior distribution.

Previously in class, we only looked at the posterior mean: $E(\theta|y)$
→ But now if θ are the parameters of interest + $h(\theta)$ is a function of θ , we can find

$$E[h(\theta|y)] = \int h(\theta) p(\theta|y) d\theta \Rightarrow E_{\text{post}}[h(\theta)] = \frac{1}{n} \sum_{i=1}^n h(\theta^{(i)})$$

→ So, just need to generate $\theta^{(1)}, \dots, \theta^{(n)}$
+ apply function $h(\theta)$ to get $h(\theta)$

→ Example

$$E(\theta|y) = \text{posterior mean, i.e. } h(\theta) = \theta^2$$

→ generate $\theta^{(1)}, \dots, \theta^{(n)}$

$$\Rightarrow \begin{pmatrix} (\theta^{(1)})^2 \\ \vdots \\ (\theta^{(n)})^2 \end{pmatrix}$$

→ approximate w/ $\frac{1}{n} \sum_{i=1}^n \theta^{(i)2}$

$$\Rightarrow E[h(\theta|y)] = E_{\text{post}}[h(\theta)] = \frac{1}{n} \sum_{i=1}^n h(\theta^{(i)})$$

→ Example → let $y_i \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2) \Rightarrow \theta = (\mu, \sigma^2)$

$$\rightarrow f(y_i | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} (y_i - \mu)^2\right\}$$

$$\rightarrow p(\underline{y} | \mu, \sigma^2) = \prod_{i=1}^n f(y_i | \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} (y_i - \mu)^2\right\}$$

$$\begin{aligned} \rightarrow D(\mu, \sigma^2) &= -2 \log[p(\underline{y} | \mu, \sigma^2)] \\ &= -2 \log\left[\prod_{i=1}^n f(y_i | \mu, \sigma^2)\right] \\ &= -2 \sum_{i=1}^n \log[f(y_i | \mu, \sigma^2)] \\ &= -2 \sum_{i=1}^n \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} (y_i - \mu)^2\right\}\right] \\ &= -2 \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y_i - \mu)^2\right] \\ &\downarrow \\ &= n \log(2\pi\sigma^2) + \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \end{aligned}$$

→ For a given dataset, $D(\mu, \sigma^2)$ is a function of μ & σ^2

5.4.2 Effective number of parameters

- This is a Bayesian measure of model complexity

$$p_D = E_{\text{post}}(D) - D(\tilde{\theta}) = 2 \left[\underbrace{\log p(y|\tilde{\theta})}_{\text{①}} - \underbrace{E_{\theta|y}[\log p(y|\theta)]}_{\text{②}} \right]$$

$E_{\text{post}}(D) = E[-2 \log p(y|\theta)]$
 $= -2 E[\log p(y|\theta)]$
 $+ D(\tilde{\theta}) = -2 \log p(y|\tilde{\theta})$

where $\tilde{\theta} = E(\theta|y)$ = posterior mean

- Estimated

$$p_D = \bar{D} - D(\hat{\theta}_{\text{Bayes}})$$

$$= 2 \left[\log p(y|\hat{\theta}_{\text{Bayes}}) - \frac{1}{S} \sum_{s=1}^S \log p(y|\theta^{(s)}) \right]$$

where $\hat{\theta}_{\text{Bayes}} = \frac{1}{S} \sum_{s=1}^S \theta^{(s)}$

Estimated posterior mean

just plug in value $\Rightarrow$ we are finding the joint pdf using the "Bayesian posterior estimate of θ "

Generate $\theta^{(1)} \rightarrow \log p(y|\theta^{(1)})$
 $\vdots \rightarrow \vdots$
 $\theta^{(S)} \rightarrow \log p(y|\theta^{(S)})$

- Another definition

$$p_{D,\text{alt}} = 2 \text{Var}_{\text{post}}[\log p(y|\theta)]$$

$\rightarrow$ use this version if the other has negatives

- Estimated

$$p_{D,\text{alt}} = 2 \frac{1}{S-1} \sum_{s=1}^S \left[\log p(y|\theta^s) - \frac{1}{S} \sum_{i=1}^S \log p(y|\theta^i) \right]^2$$

- Comparison:

- p_D is more stable.
- $p_{D,\text{alt}}$ is always positive.

Think of this as the same random variable
 $\Rightarrow p_{D,\text{alt}} \approx \frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})^2$
 sample variance

5.4.3 DIC

$$\text{DIC} = D(\tilde{\theta}) + 2p_D = -2 \log p(y|\tilde{\theta}) + 2p_D$$

$\rightarrow$ Another quantity

$$\rightarrow \text{AIC} = -2 \log p(y|\tilde{\theta}) + 2p_0$$

$\rightarrow$ Another form

$\rightarrow$ under the first definition if $p_0 = E_{\text{post}}(D) - D(\tilde{\theta})$

$$\begin{aligned} \Rightarrow \text{DIC} &= D(\tilde{\theta}) + (p_0 + p_0) \\ &= \downarrow + [E_{\text{post}}(D) - D(\tilde{\theta}) + p_0] \\ &= E_{\text{post}}(D) + p_0 \\ &= \downarrow + (E_{\text{post}}(D) - D(\tilde{\theta})) \\ &= 2 E_{\text{post}}(D) - D(\tilde{\theta}) \end{aligned}$$

$$D(\tilde{\theta}) = -2 \log p(y|\tilde{\theta})$$

$\hookrightarrow$ just plug in value of posterior mean

$$\text{DIC} = \underbrace{-2 \log p(y|\tilde{\theta})}_{\text{How well model fits data}} + \underbrace{2p_0}_{\text{penalty}}$$

- Estimate $\rightarrow \text{DIC} = -2 \log p(y|\tilde{\theta}) + 2p_0$

$$\downarrow = \text{①} + \text{②}$$

$$\rightarrow \text{estimates} \rightarrow \text{①} \Rightarrow -2 \log p(y|\hat{\theta}_{\text{Bayes}})$$

$\hat{\theta}_{\text{Bayes}} = \frac{1}{M} \sum_{m=1}^M \theta^{(m)}$ from posterior samples

$$\rightarrow \text{②} \Rightarrow 2[\bar{D} - D(\hat{\theta}_{\text{Bayes}})]$$

$$\downarrow$$

$$D^{(m)} = -2 \sum_{i=1}^n \log p(y_i|\theta^{(m)})$$

$$\Rightarrow \bar{D} = \frac{1}{M} \sum_{m=1}^M D^{(m)}$$

- For almost all information criterion, models with smaller values are preferred

- Rule of Thumb

- If difference of DIC is 10 or higher, the model with smaller DIC is selected
- Difference between 5 and 10 is substantial
- If difference is below 5, no real difference between the two models

(Nothing to do w/ posterior mean)

use lppd package in R
 → preferred to DSE bc it can be found using a package in R rather than manually

5.4.4 Watanabe-Akaike or widely applicable information criterion (WAIC)

- $\text{lppd} = \log \text{pointwise predictive density} = \log \prod_{i=1}^n p_{\text{post}}(y_i) = \sum_{i=1}^n \log \int p(y_i | \theta) p_{\text{post}}(\theta) d\theta = \sum_{i=1}^n \log [E_{\text{post}}(p(y_i | \theta))]$

↳ measure of how well the model fits the data

↳ $p(y_i | \theta)$ is the pdf (pdf) of the i th observation, y_i
 → it is a function of θ

↳ $\int p(y_i | \theta) p_{\text{post}}(\theta) d\theta = E_{\text{post}}[p(y_i | \theta)]$
 ↳ posterior dist of θ

- Computed $\hat{\text{lppd}}$

$$\rightarrow E_{\text{post}}[p(y_i | \theta)] \xrightarrow{\text{estimate}} \frac{1}{M} \sum_{m=1}^M p(y_i | \theta^{(m)})$$

$$\Rightarrow \hat{\text{lppd}} = \sum_{i=1}^n \log \left[\frac{1}{M} \sum_{m=1}^M p(y_i | \theta^{(m)}) \right]$$

- Defined $p_{\text{WAIC}1}$ as

Based on expected value

$$p_{\text{WAIC}1} = 2 \sum_{i=1}^n (\log E_{\text{post}} p(y_i | \theta) - E_{\text{post}} (\log (y_i | \theta)))$$

Just replace these w/ the mean of the samples

– Estimated:

$$\rightarrow 2 \sum_{i=1}^n \log \left[\frac{1}{M} \sum_{m=1}^M p(y_i | \theta^{(m)}) \right] - \frac{1}{M} \sum_{m=1}^M \log [y_i | \theta^{(m)}]$$

→ $\theta^{(m)}$ are of course from the posterior samples

- Defined $p_{\text{WAIC}2}$ as

Based on variance

$$p_{\text{WAIC}2} = \sum_{i=1}^n \text{var}_{\text{post}} (\log p(y_i | \theta))$$

– Estimated: → use sample variance of $\log [p(y_i | \theta^{(m)})]$

$$\rightarrow V_i = \frac{1}{M-1} \sum_{m=1}^M [\log [p(y_i | \theta^{(m)})] - \log [p(y_i | \theta^{(m)})]^2]$$

$$\rightarrow \text{Estimate of } \text{var}_{\text{post}} [\log p(y_i | \theta)] = \sum_{i=1}^n V_i$$

- Then WAIC is defined as

$$\text{WAIC} = -2\text{lppd} + 2p_{\text{WAIC}}$$

- Usually $p_{\text{WAIC}2}$ is preferred, closer to leave-one-out.

multiply by these constants so that its form is similar to AIC & BIC

→ Another measure → $\text{elpd} = -\frac{1}{2} \text{WAIC}$
 ↓
 = $\text{lppd} - p_{\text{WAIC}}$